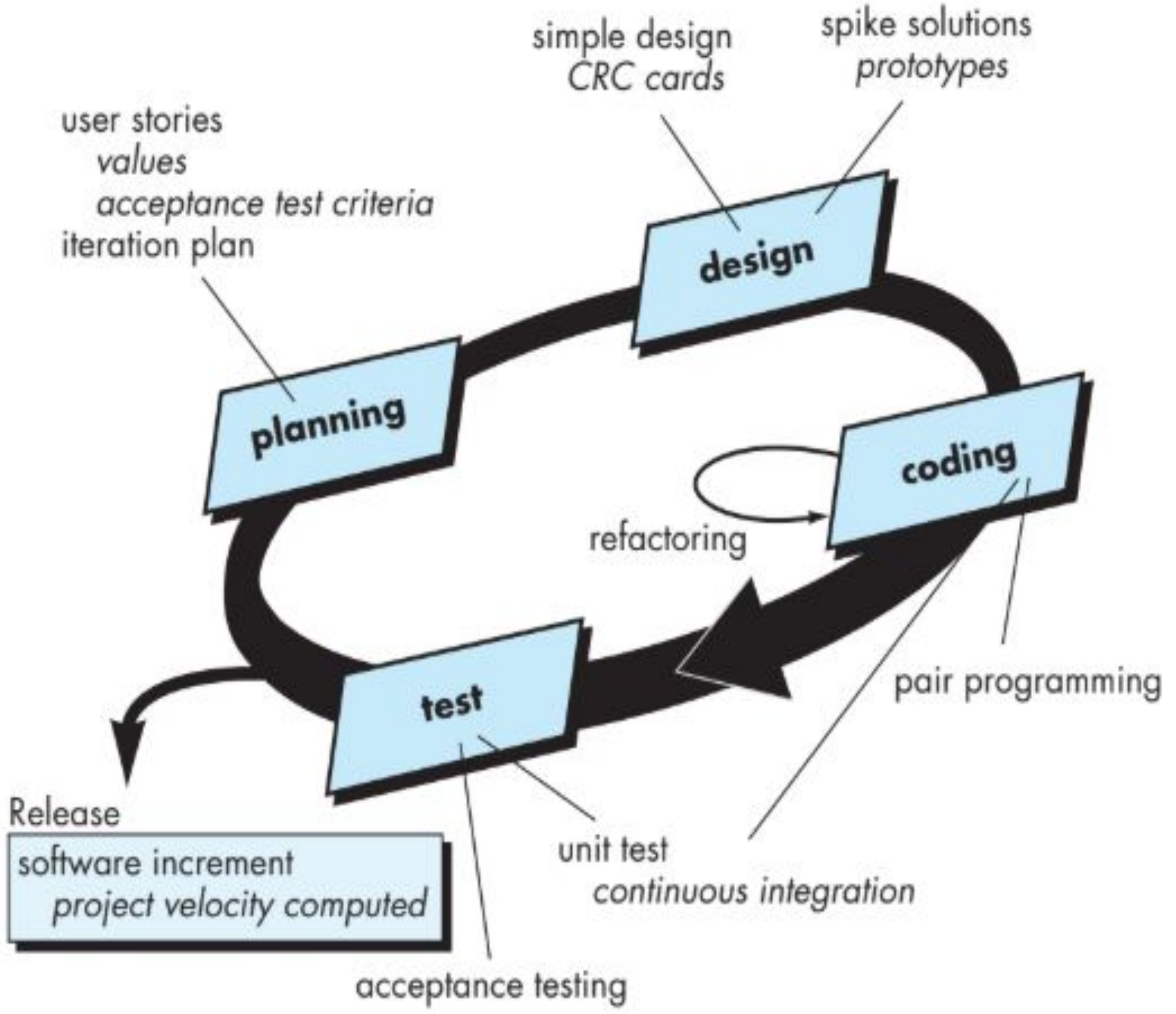
XP(Extreme Programming)

XP

- Extreme Programming (XP) uses an object-oriented approach in software development. In this approach, the system is designed using classes and objects that represent real-world entities. XP promotes small, simple classes with single responsibilities. Object-oriented concepts such as encapsulation, inheritance, and polymorphism help in improving code readability and reusability. Since XP encourages frequent refactoring and Test-Driven Development, object-oriented design makes it easier to modify and test the code without affecting the entire system. Thus, the object-oriented approach supports flexibility, maintainability, and high-quality software in XP.

The Extreme Programming process



Planning

- Planning. The planning activity begins with listening—a requirements gathering activity that enables the technical members of the XP team to understand the business context for the software and to get abroad feel for required output and major features and functionality.
- User stories -required output, features, and functionality for software to be built.
- Each story is written by the customer and is placed on an index card. The customer assigns a value (i.e., a priority) to the story

Planning(contd.)

- Members of the XP team then assess each story and assign a cost—measured in development weeks—to it.
- If the story >3 weeks, the customer is asked to split the story into smaller stories and the assignment of value and cost occurs again.
- Customers and developers work together to decide how to group stories into the next release (the next software increment) to be developed by the XP team.

Planning(contd.)

- The XP team orders the stories that will be developed in one of three ways:

(1) all stories will be implemented immediately (within a few weeks),

(2) the stories with highest value will be moved up in the schedule and implemented first, or

(3) the riskiest stories will be moved up in the schedule and implemented first.

Planning(contd.)

- Project velocity is the number of customer stories implemented during the first release. Project velocity can then be used to

(1) help estimate delivery dates and schedule for subsequent releases and

(2) determine whether an over commitment has been made for all stories across the entire development project. If an over commitment occurs, the content of releases is modified or end delivery dates are changed.

Design :KIS (keep it simple) principle.

- XP encourages the use of CRC cards(class-responsibility-collaborator)
- For a difficult design problem, XP recommends the immediate creation of an operational prototype of that portion of the design. called a spike solution, the design prototype is implemented and

evaluated. The intent is to lower risk when true implementation starts and to validate the original estimates for the story containing the design problem.

Refactoring

- Refactoring is the process of changing a software system in such a way that it does not alter the external behaviour of the code yet improves the internal structure. It is a disciplined way to clean up code [and modify/simplify the internal design] that minimizes the chances of introducing bugs. In essence, when you refactor you are

  improving the design of the code after it has been written.

Refactoring means that design occurs continuously as the system is constructed.

Coding

- After stories are developed and preliminary design work is done, the team does not move to code, but rather develops a series of unit tests that will exercise each of the stories that is to be included in the current release (software increment).
- Once the unit test has been created, the developer is better able to focus on what must be implemented to pass the test.

Pair programming

- XP recommends that two people work together at one computer workstation to create code for a story.
- As pair programmers complete their work, the code they develop is integrated with the work of others.
- Once the code is complete, it can be unit-tested immediately, thereby providing instantaneous feedback to the developers.

Testing

- XP acceptance tests, also called customer tests, are specifi ed by the customer and focus on overall system features and functionality that are visible and reviewable by the customer. Acceptance tests are derived from user stories that have been implemented as part of a software release.